PROGRAMMING ASSIGNMENT 1: WORD SIMILARITY AND SYNONYM-ANTONYM CLASSIFICATION

(*Working in group of 3 students; Deadline: 9/10; Submitting via the course website*)

A. Problem

1. *Cosine similarity*: Given pre-trained embeddings of Vietnamese words, implement a function for calculating cosine similarity between word pairs. Test your program using word pairs in ViSim-400 dataset (in directory Datasets/ViSim-400).
2. *K-nearest words*: Given a word w, find k most-similar words of w using the function implemented in question 1.
3. *Synonym-antonym classification*: Implement a classifier (Logistic Regression, Multi-layer Perceptron, …) for distinguishing synonyms and antonyms. For training, you are given a data set in directory antonym-synonym set: *https://github.com/NLP-Projects/Word-Similarity/tree/master/antonym-synonym%20set*

For testing, you use ViCon-400 (in directory Datasets/ViCon-400). Experimental results are evaluated by precision scores, recall scores and F-measure (F1).

$$Precision = \frac{True\ positive}{True\ positive + False\ positive}$$

$$Recall = \frac{True\ positive}{True\ positive + False\ negative}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

B. Programming language and tool

1. Programming language: Python.
2. Machine learning library: Scikit-learn (https://scikit-learn.org/).

C. Data sets

Available at: ***https://github.com/NLP-Projects/Word-Similarity***

1. Pre-trained Word2Vec in directory *Word2vec*.

For example, the word "*sinh_viên*" is represented by a 150-dimension vector as follows:

sinh_viên -0.16830535 -0.46649584 -0.09095726 0.26220384 -0.06665505 0.031103149 -0.09404702 0.04053062 0.11192394 -0.5279207 0.14246537 0.001038614 -0.35036477 -0.13038214 -0.09090571 0.16573012 -0.36303198 0.09883489 0.18725859 0.13902836 0.06649695 -0.084366314 -0.0027223954 -0.19726162 0.2759428 0.19625933 -0.05050645 -0.2912832 0.3332136 -0.05097548 -0.033659954 0.08438698 0.08262109 0.17202266 0.028343566 -0.2239018 -0.12655334 -0.007960333 -0.11457155 0.05343645 0.04817521 -0.07958502 0.14042595 -0.03268163 0.15085939 -0.019562919 -0.3306483 -0.12484468 0.28407142 0.412389 -0.20573528 0.1886385 -0.32018003 -0.28494123 0.30086038 -0.16913833 -0.15604256 0.014038047 0.0027964886 -0.042381432 0.10412829 -0.25038096 -0.1707933 0.28154066 -0.18622164 -0.12197793 -0.3312451 -0.51899064 -0.30050278 -0.25337204 0.0281554 0.05594583 0.2137868 -0.063191235 -0.25344792 -0.041951556 -0.24166772 -0.06607595 -0.0058164373 0.20314898 0.22446826 -0.27940097 -0.20987804 0.393391 0.19083196 0.1140723 -0.0413766 0.2983006 0.09309805 -0.014998496 0.56122595 -0.21278302 -0.29513258 -0.5796372 -0.104330115 -0.049805988 -0.20145701 0.07974479 -0.08291912 0.20195685 0.5489658 0.27150062 0.28475645 0.047555167 -0.05718565 -0.25075287 0.16845582 -0.26866978 0.24444789 -0.1013144 0.34333876 0.44675854 0.2024813 -0.18543775 -0.428579 0.00044292794 -0.06001611 0.1679784 -0.18539493 0.5264743 0.032929808 -0.11656791 0.11542175 0.345688 -0.16840588 -0.10268665 0.16477033 -0.2467253 0.1251898 0.0076962546 -0.27712318 0.33789232 -0.1583204 0.19434617 -0.23928708 0.177563 0.11510917 0.2388904 -0.50243086 -0.53305346 0.01388552 0.18508705 0.28942367 0.10520081 -0.060460173 0.16218448 0.13140707 -0.16748051 -0.10331021 -0.17137058

2. VSim-400 dataset in directory *Datasets/ViSim-400*.

Line format and examples (Sim1 represents human rating of similarity in the interval [0,4], Sim2 represents human rating of similarity in the interval [0,6]):

Word1	Word2	POS	Sim1	Sim2	STD
biến	ngập	V	3.13	5.22	0.72
động	tĩnh	V	0.6	1.0	0.95
khuyết	ưu	N	0.2	0.33	0.4
cõi_tục	cõi_âm	N	0.6	1.0	0.95
kết_duyên	thành_hôn	V	5.27	8.78	1.06
cấp_tiến	bảo_thủ	A	0.87	1.45	1.15
nước_lớn	nguy_hiểm	N	1.07	1.78	1.12
bất_lợi	thuận_lợi	N	0.33	0.55	0.79
phân_ly	sum_họp	V	0.47	0.78	1.09
diễu_hành	tuần_hành	V	4.53	7.55	1.15

3. Vcon-400 dataset in directory *Datasets/ViCon-400*.

Line format and examples:

Word1	Word2	Relation
hời_hợt	nông_cạn	SYN
thảnh_thơi	ưu_tư	ANT
vô_lý	có_lí	ANT
cuồng_nộ	phẫn_nộ	SYN
đần	thông_minh	ANT
có_lí	vô_lí	ANT
ấm_no	no_ấm	SYN
lí_thú	lý_thú	SYN

4. Reference documents in directory *Reference*.